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Jain et al.

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(54) **HIGH ELECTRON MOBILITY TRANSISTOR DEVICES HAVING A SILICIDED POLYSILICON LAYER**

(71) Applicant: **GlobalFoundries U.S. Inc.**, Malta, NY (US)

(72) Inventors: **Vibhor Jain**, Clifton Park, NY (US); **Johnatan Avraham Kantarovsky**, South Burlington, VT (US); **Mark David Levy**, Williston, VT (US); **Ephrem Gebreselasie**, South Burlington, VT (US); **Yves Ngu**, Birchwood Drive Hinesburg, VT (US); **Siva P. Adusumilli**, South Burlington, VT (US)

(73) Assignee: **GlobalFoundries Singapore Pte. Ltd.**, Singapore (SG)

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See application file for complete search history.

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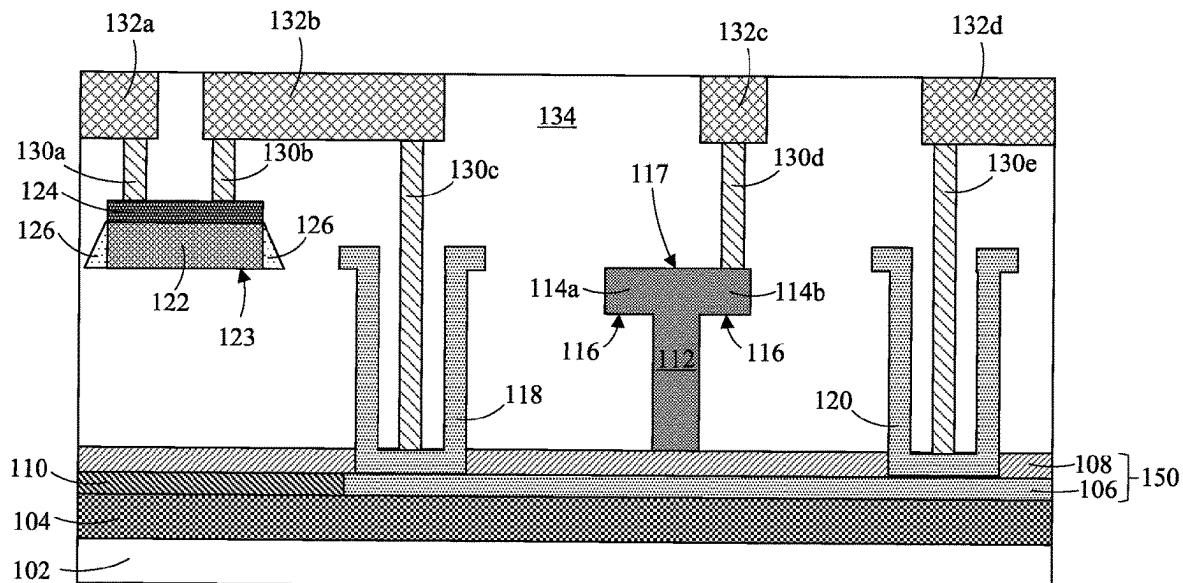
Primary Examiner — Mohammad A Rahman

(74) *Attorney, Agent, or Firm* — Francois Pagette

(57) **ABSTRACT**

The present disclosure relates generally to structures in semiconductor devices and methods of forming the same. More particularly, the present disclosure relates to high electron mobility transistor (HEMT) devices having a silicided polysilicon layer. The present disclosure may provide an active region above a substrate, source and drain electrodes in contact with the active region, a gate above the active region, the gate being laterally between the source and drain electrodes, a polysilicon layer above the substrate, and a silicide layer on the polysilicon layer. The active region includes at least two material layers with different band gaps. The polysilicon layer may be configured as an electronic fuse, a resistor, or a diode.

19 Claims, 13 Drawing Sheets



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H10D 64/60 (2025.01)
H10D 64/66 (2025.01)
- (52) **U.S. Cl.**
 CPC *H10D 64/605* (2025.01); *H10D 64/661*
 (2025.01); *H10D 64/671* (2025.01)

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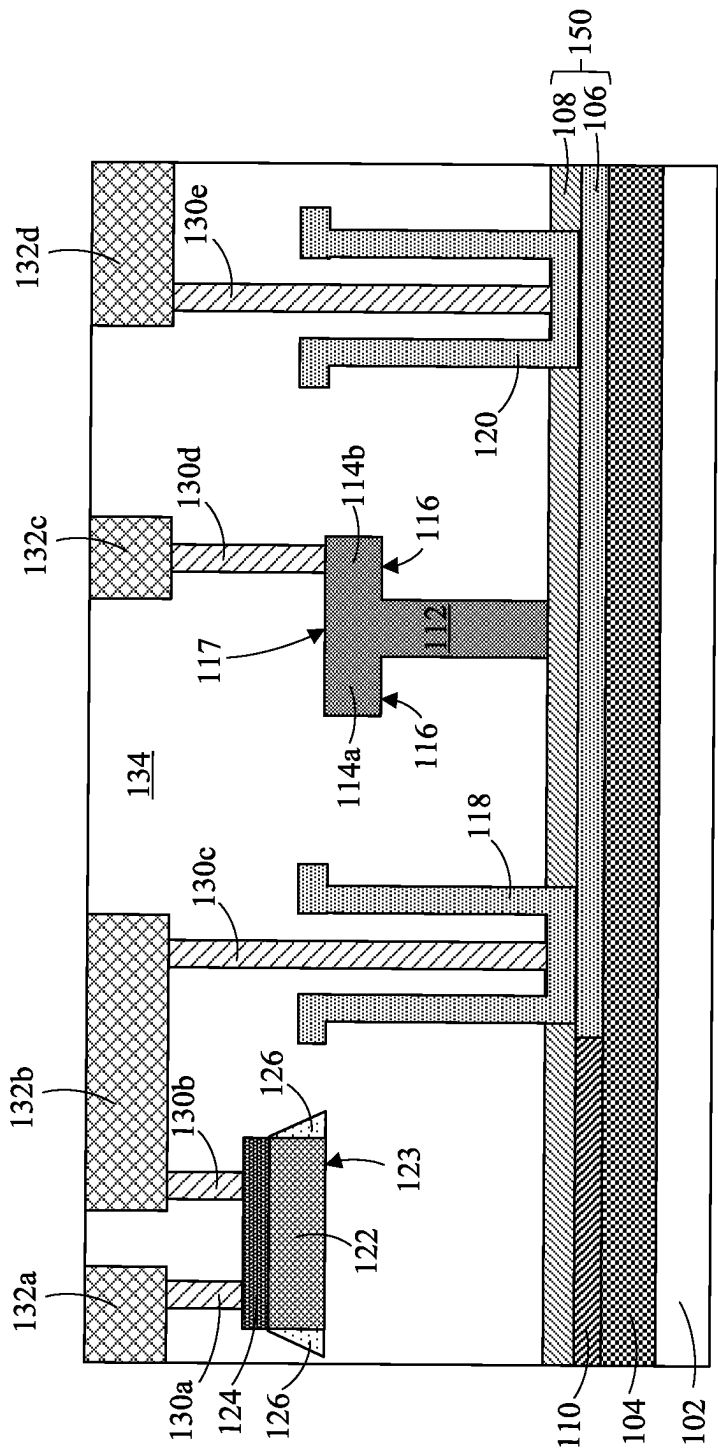


FIG. 1

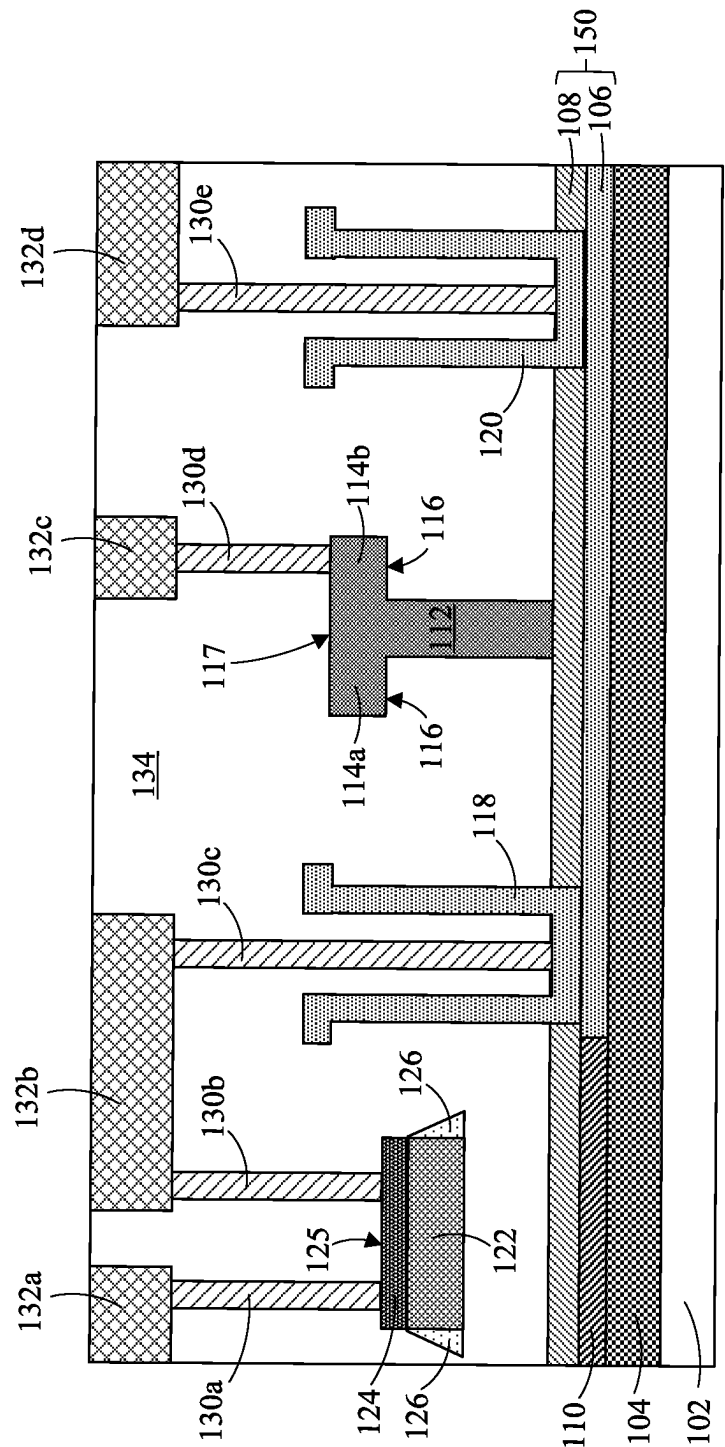


FIG. 2

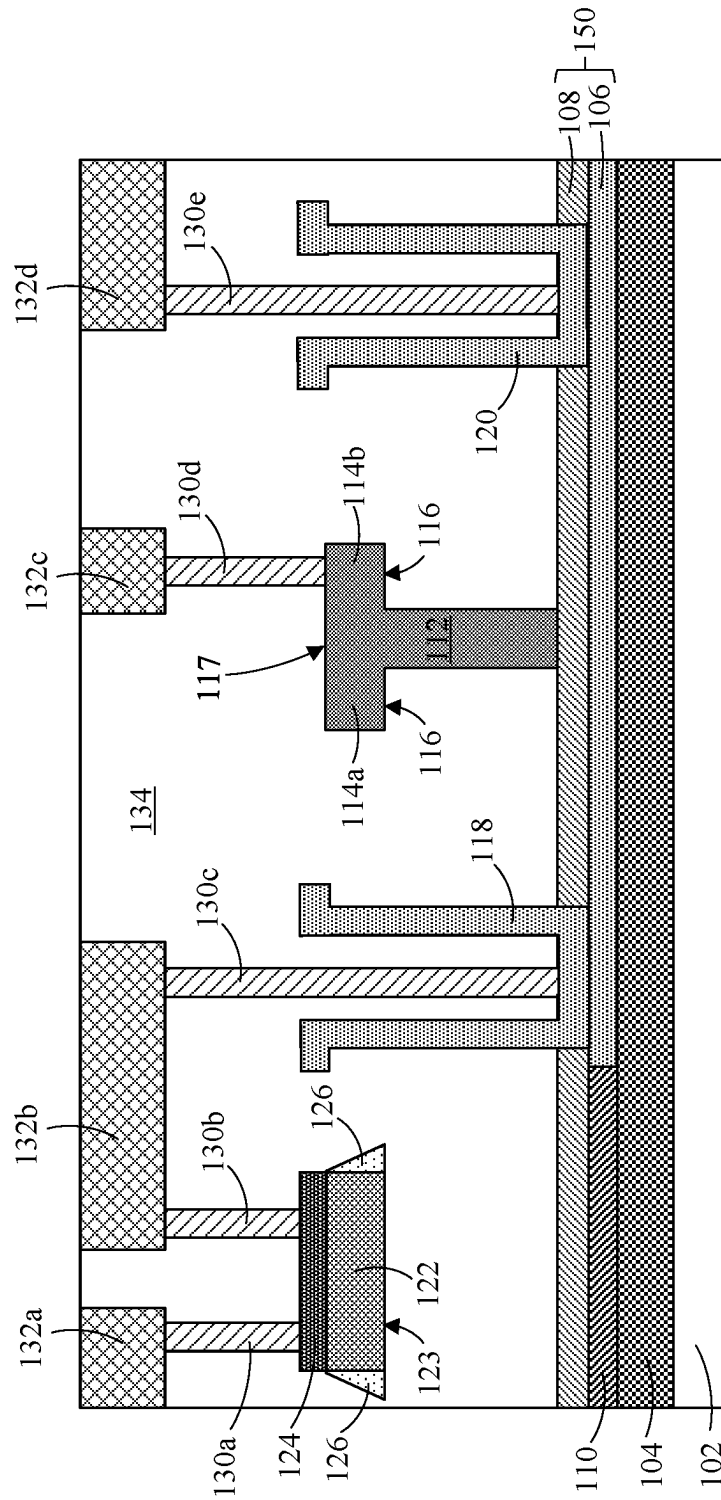


FIG. 3

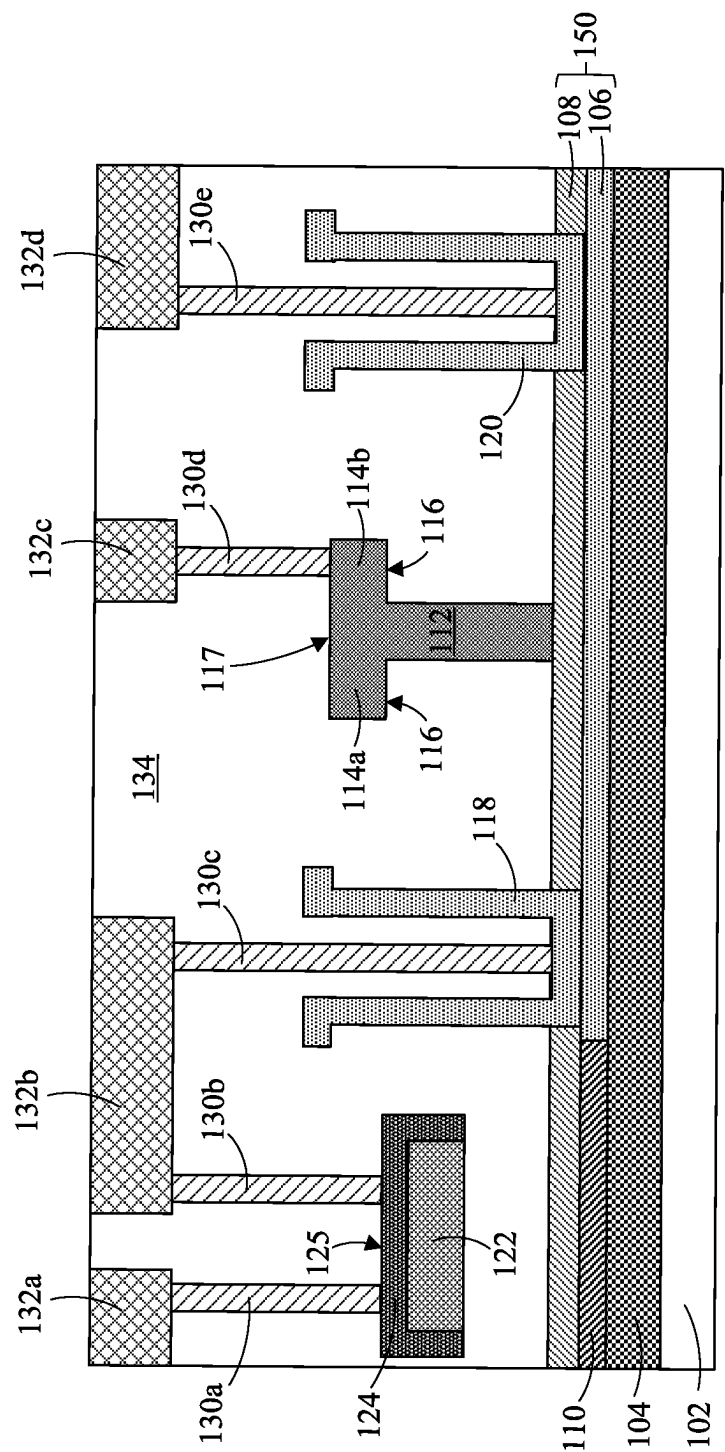


FIG. 4

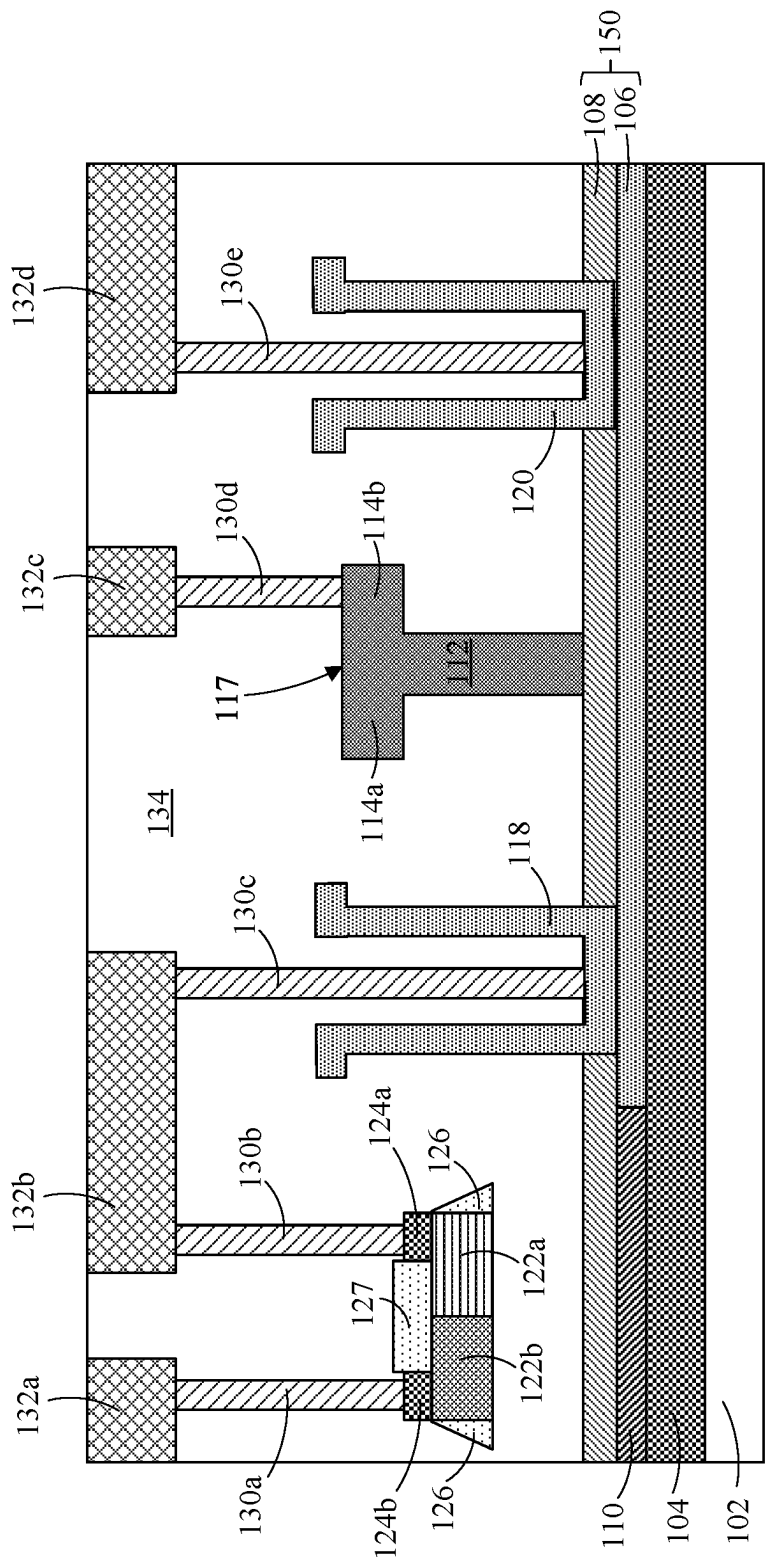


FIG. 5

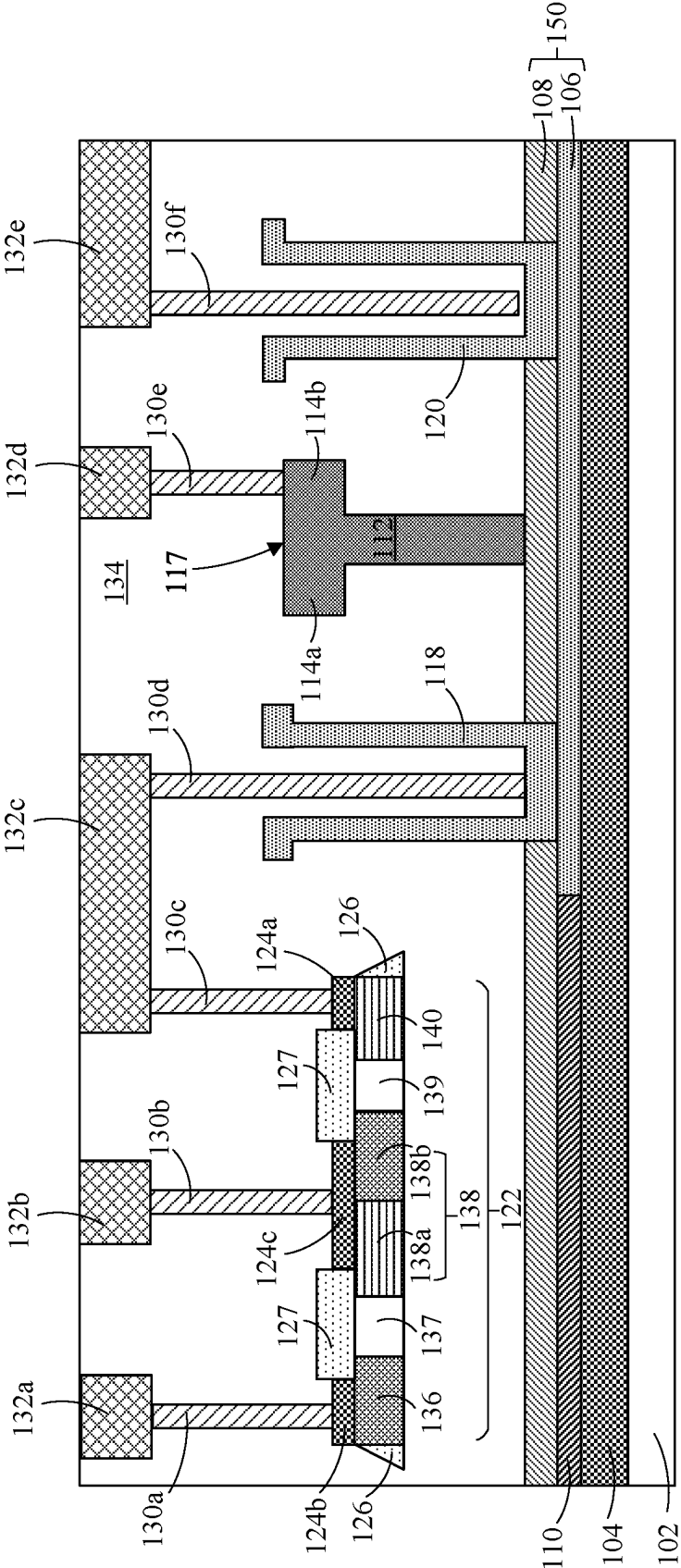


FIG. 6

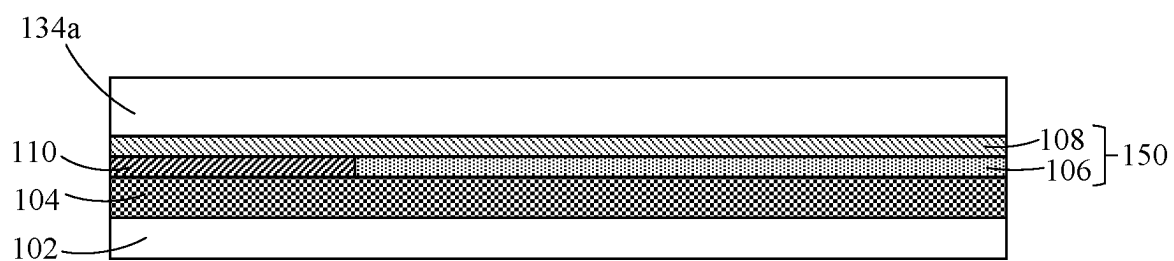


FIG. 7

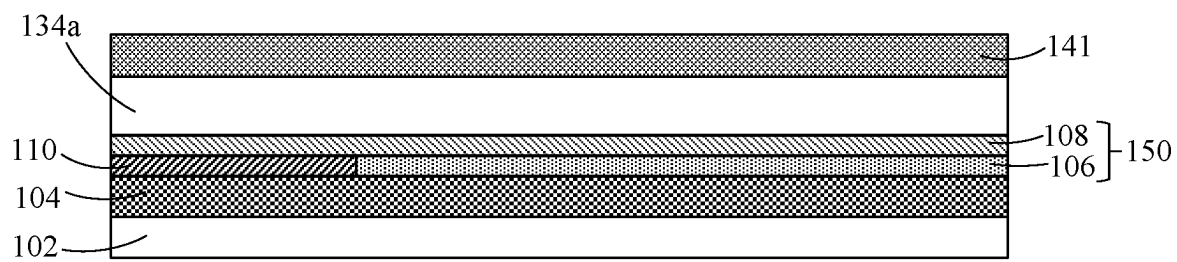


FIG. 8

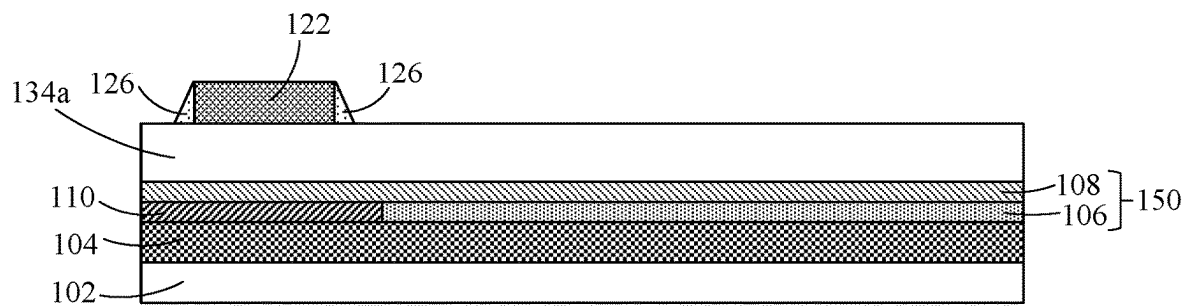


FIG. 9A

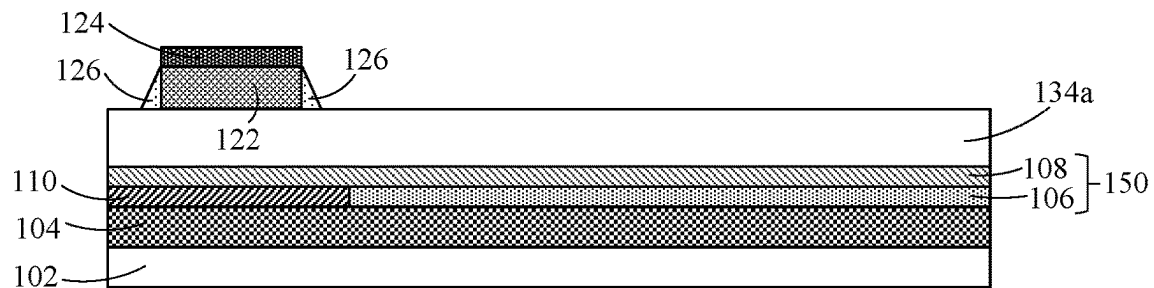


FIG. 9B

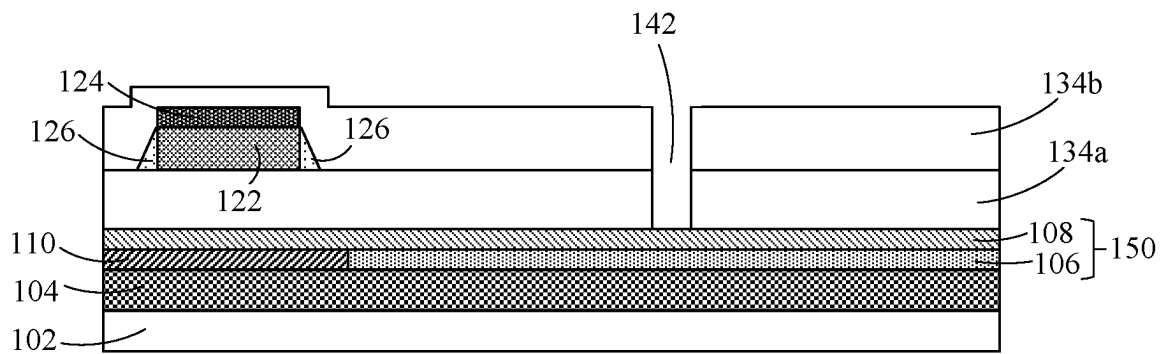


FIG. 10

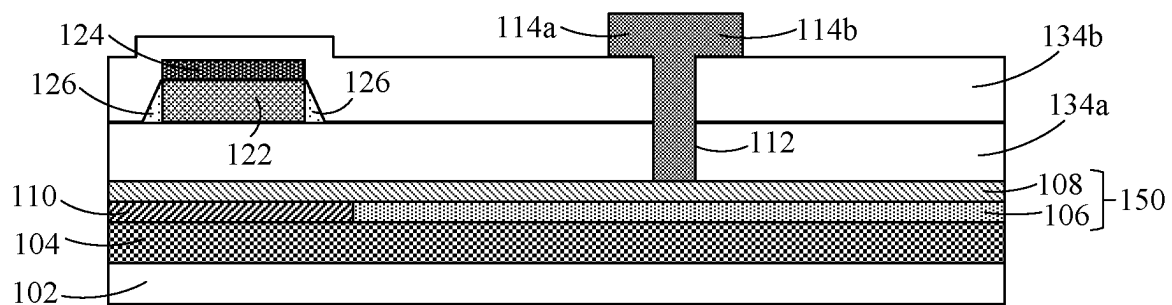


FIG. 11

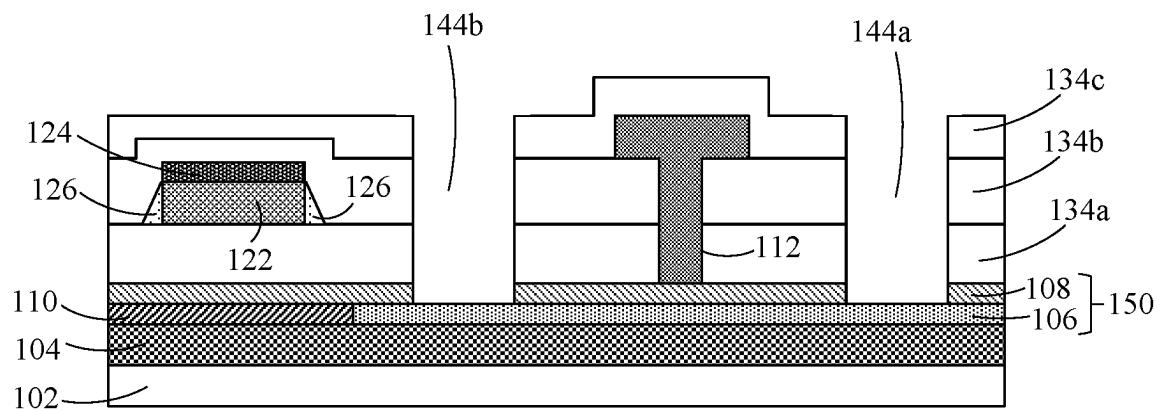


FIG. 12

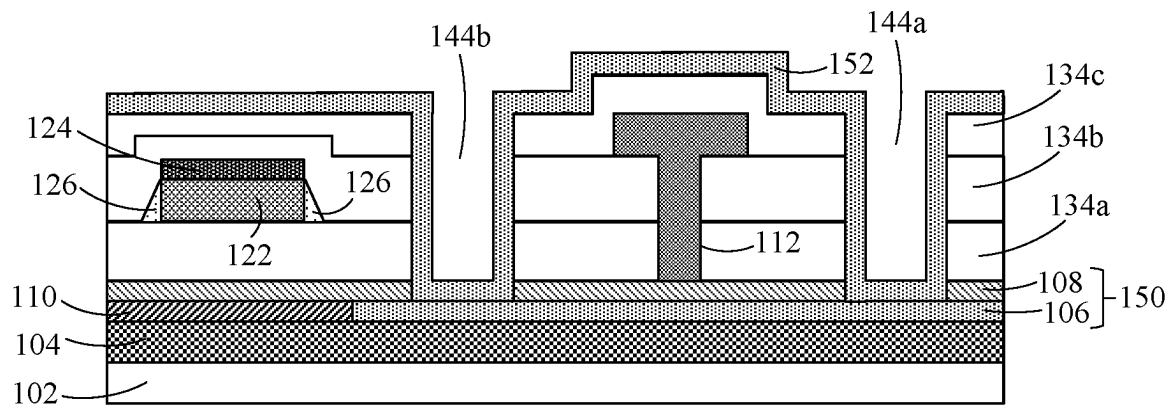


FIG. 13

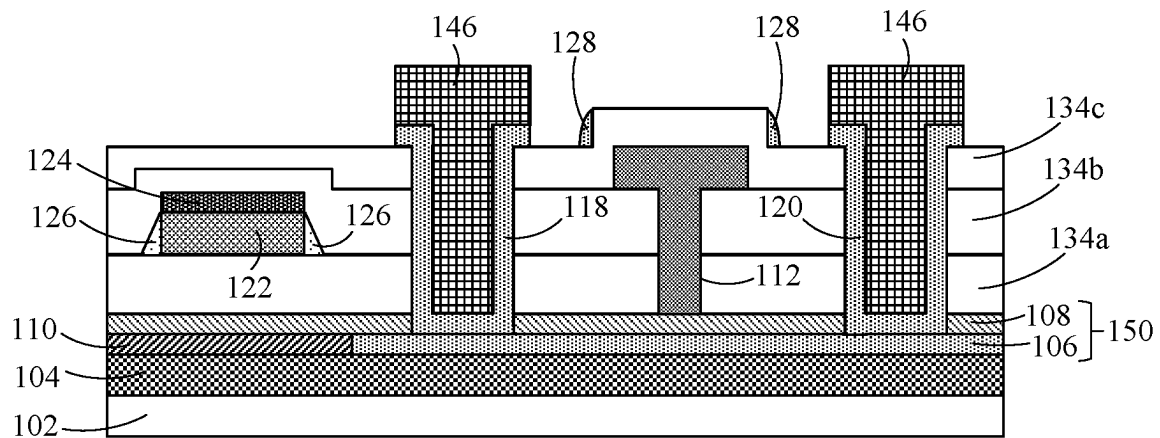


FIG. 14

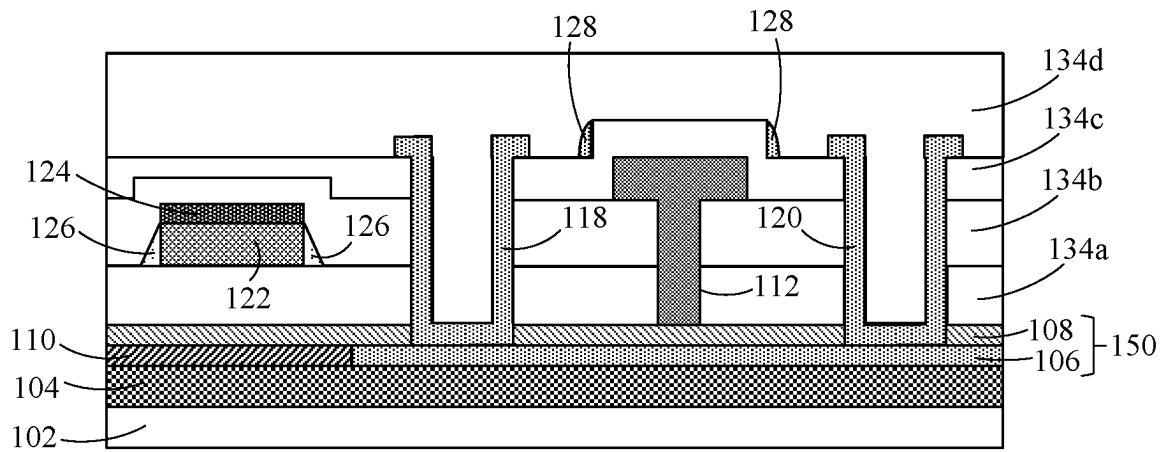


FIG. 15

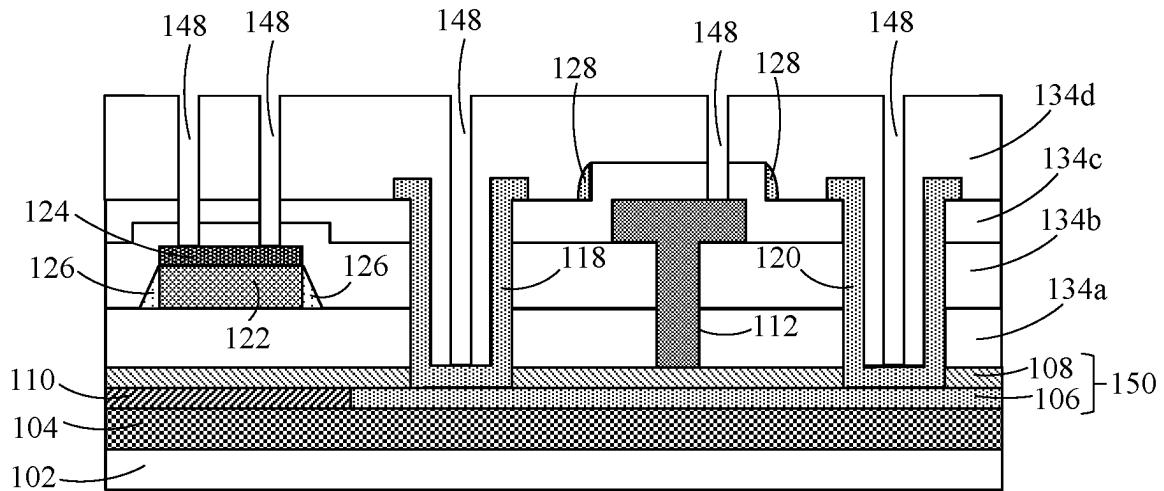


FIG. 16

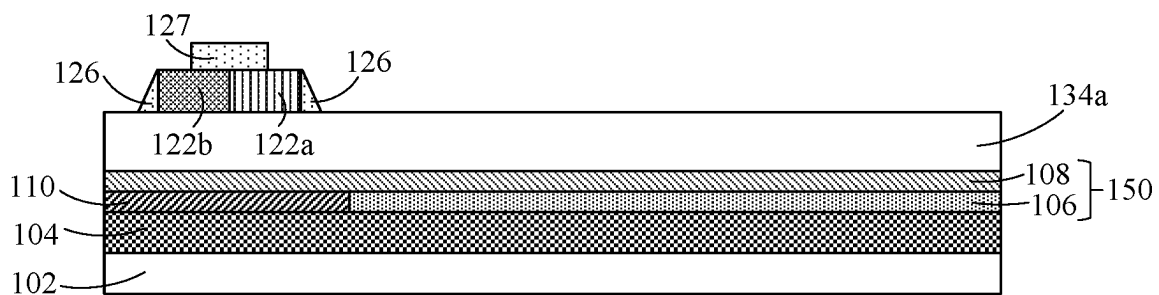


FIG. 17A

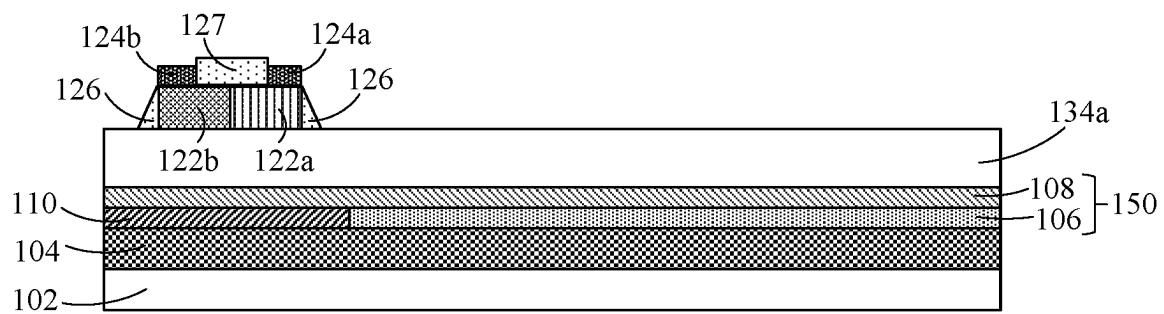


FIG. 17B

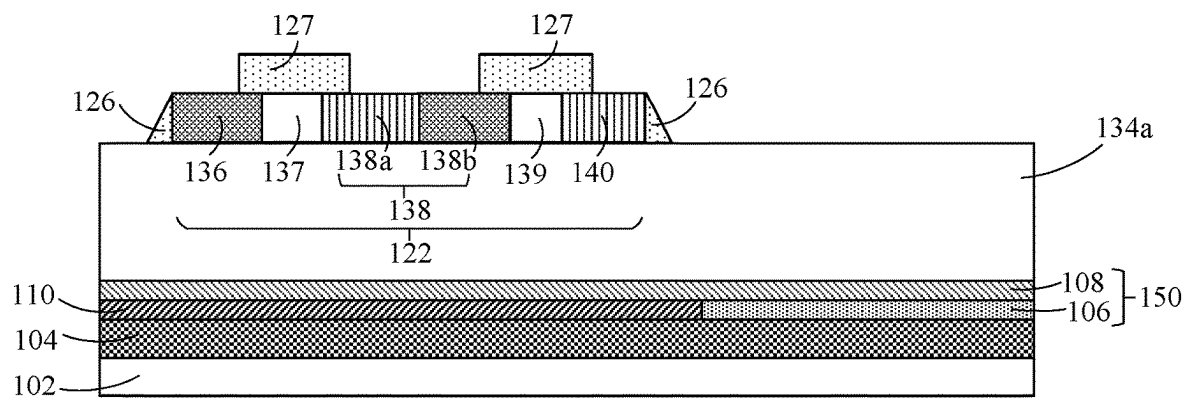


FIG. 18A

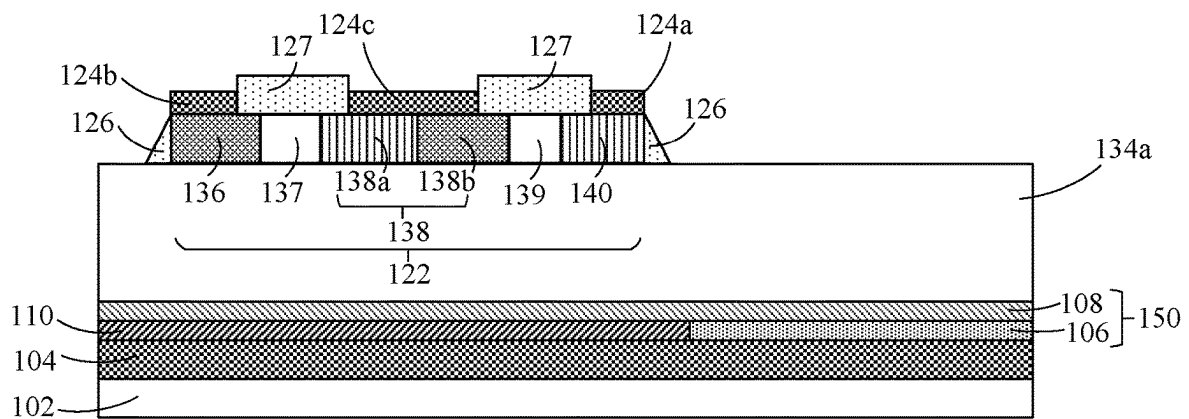


FIG. 18B

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HIGH ELECTRON MOBILITY TRANSISTOR DEVICES HAVING A SILICIDED POLYSILICON LAYER

FIELD OF THE INVENTION

The present disclosure relates generally to structures in semiconductor devices and methods of forming the same. More particularly, the present disclosure relates to high electron mobility transistor (HEMT) devices having a silicided polysilicon layer.

BACKGROUND

High electron mobility transistors (HEMT) have become increasingly desirable due to their ability to switch at high frequency, carry large currents, and support high voltages. HEMT devices can typically withstand high voltages (e.g., 600V to 1200V) while operating at high frequencies (e.g., 100 kHz to 100 GHz). Integrated circuit chips having the HEMT devices may also include additional components, such as active or passive components. Examples of these active or passive components can include fuses, anti-fuses, resistors, or diodes. The HEMT device in the integrated circuit chip can be used to program those additional components.

Fuses and anti-fuses are widely used in field programmable devices. An example of a fuse includes an electrically programmable fuse (or an "electronic fuse"). An electronic fuse may operate based on electromigration, where current crowding takes place around a fixed location, thus initiating the electromigration which results in further current crowding and material migration in the direction of the electron movement along the fuse element. This may result in an increase in a few orders of magnitude in the electronic fuse's resistance.

However, for device applications that utilize a HEMT to program the electronic fuse, the HEMT and the electronic fuse may be built on separate dies or wafers. The fabrication of the HEMT and the electronic fuse on different dies or wafers may incur additional manufacturing costs, as well as causing the fabricated chip to have a larger device size and area.

SUMMARY

In an aspect of the present disclosure, there is provided a semiconductor device including an active region above a substrate, the active region including at least two material layers with different band gaps, source and drain electrodes in contact with the active region, a gate above the active region, the gate is laterally between the source and drain electrodes, the gate has an upper surface, a polysilicon layer above the substrate, the polysilicon layer has a lower surface, in which the polysilicon layer is positioned laterally adjacent to the gate and the lower surface of the polysilicon layer is substantially coplanar with the upper surface of the gate, and a silicide layer on the polysilicon layer.

In another aspect of the present disclosure, there is provided a semiconductor device including a buffer layer on a substrate, an active region above the substrate, the active region having a channel layer disposed on the buffer layer, and a barrier layer disposed on the channel layer, source and drain electrodes in contact with the channel layer, a gate on the barrier layer, the gate is laterally between the source and drain electrodes, the gate has an upper surface, a polysilicon layer above the barrier layer, the polysilicon layer has a

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lower surface, the polysilicon layer is positioned laterally adjacent to the gate, and the lower surface of the polysilicon layer is substantially coplanar with the upper surface of the gate, in which the polysilicon layer includes heavily doped regions and lightly doped regions, and a silicide layer on each of the heavily doped regions.

In yet another aspect of the present disclosure, there is provided a method of forming a semiconductor device by forming an active region above a substrate, the active region including at least two material layers with different band gaps, forming source and drain electrodes to contact the active region, forming a gate above the active region, the gate is laterally between the source and drain electrodes, the gate has an upper surface, forming a polysilicon layer above the substrate, the polysilicon layer has a lower surface, in which the polysilicon layer is positioned laterally adjacent to the gate and the lower surface of the polysilicon layer is substantially coplanar with the upper surface of the gate, and forming a silicide layer on the polysilicon layer.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure may be understood by reference to the following description taken in conjunction with the accompanying drawings.

For simplicity and clarity of illustration, the drawings illustrate the general manner of construction, and certain descriptions and details of features and techniques may be omitted to avoid unnecessarily obscuring the discussion of the described embodiments of the present disclosure. Additionally, elements in the drawings are not necessarily drawn to scale. For example, the dimensions of some of the elements in the drawings may be exaggerated relative to other elements to help improve understanding of embodiments of the present disclosure. The same reference numerals in different drawings denote the same elements, while similar reference numerals may, but do not necessarily, denote similar elements.

FIG. 1 through FIG. 6 are cross-sectional views of examples of semiconductor devices.

FIG. 7, FIG. 8, FIG. 9A, FIG. 9B, FIG. 10 through FIG. 16 are cross-sectional views depicting structures at various stages of forming a semiconductor device shown in FIG. 2.

FIG. 17A and FIG. 17B are cross-sectional views depicting structures at stages subsequent to FIG. 7 for forming a semiconductor device shown in FIG. 5.

FIG. 18A and FIG. 18B are cross-sectional views depicting structures at stages subsequent to FIG. 7 for forming a semiconductor device shown in FIG. 6.

DETAILED DESCRIPTION

Various illustrative embodiments of the present disclosure are described below. The embodiments disclosed herein are exemplary and not intended to be exhaustive or limiting to the present disclosure.

Referring to FIG. 1, FIG. 2, FIG. 3, and FIG. 4, there are provided examples of a semiconductor device including a substrate 102, an active region 150 above the substrate 102, a source electrode 120 in contact with the active region 150, a drain electrode 118 in contact with the active region 150, a gate 112 above the active region 150, a polysilicon layer 122 above the substrate 102, and a silicide layer 124 on the polysilicon layer 122. The gate 112 is positioned laterally between the source electrode 120 and the drain electrode 118. The polysilicon layer 122 may be electrically connected to either the drain electrode 118 or the source electrode 120.

It should be noted that the relative positions of the drain electrode **118** and the source electrode **120** as illustrated in the accompanying drawings may be interchangeable. For example (as shown), the drain electrode **118** may be positioned laterally between the gate **112** and the polysilicon layer **122**. Alternatively (not shown), the source electrode **120** may be positioned laterally between the gate **112** and the polysilicon layer **122**. Although not shown in the accompanying drawings, the polysilicon layer **122** may alternatively be electrically connected to the gate **112**. As used herein, the term “active region” may refer to a region that provides a pathway for the drifting of electrical charges between the source electrode **120** and the drain electrode **118**.

The substrate **102** may include any semiconductor material, such as silicon, germanium, silicon germanium (SiGe), silicon carbide (SiC), and those consisting essentially of III-V compound semiconductors, such as gallium arsenide (GaAs), gallium nitride (GaN), II-VI compound semiconductors such as zinc selenide (ZnSe). Alternatively, the substrate **102** may include sapphire, spinel, zinc oxide (ZnO), aluminum nitride (AlN), or any other material capable of supporting the growth of Group III-nitride materials.

The substrate **102** may be a bulk substrate as illustrated in the accompanying drawings. In alternative embodiments (not shown), the substrate **102** may be a composite substrate structured to have multiple layers of the materials as described hereinbefore. Examples of the composite substrate may include, but are not limited to, a silicon-on-ceramic substrate, Si/SiGe layered substrate, a silicon-on-insulator (SOI) substrate, a germanium-on-insulator (GOI) substrate, or a SiGe-on-insulator substrate.

The gate **112** may modulate the conductance of the active region **150** and control the drifting of electric charges between the source electrode **120** and the drain electrode **118**. The gate **112** may include a metallic material such as, but not limited to, tungsten, cobalt, nickel, copper, and aluminum. In some embodiments, the gate **112** may have field plate extensions **114a**, **114b** that extend laterally from the gate **112** towards the source **120** and drain **118** electrodes. The field plate extensions **114a**, **114b** may be integrally formed with the gate **112**. The gate **112** with the field plate extensions **114a**, **114b** may be referred to as a “T-shaped” gate or “F-shaped” gate. The inclusion of the field plate extensions **114a**, **114b** may increase the breakdown voltage of the device. The gate **112** may have an upper surface **117** and each field plate extension **114a**, **114b** may have a lower surface **116**. Alternatively, in other embodiments (not shown), the gate **112** does not have the field plate extensions **114**, **114b**.

The conductance of the active region **150** may be controlled by the amount of voltage applied to the gate **112**. The active region **150** may have various structural configurations depending on the desired type of transistor. The active region **150** includes at least two material layers with different band gaps. In particular, the active region **150** may include a heterostructure having multiple layers of materials. For example, the active region **150** may include a barrier layer **108** disposed on a channel layer **106**.

The channel layer **106** may include one or more sub-layers with a desired band gap. The sub-layers may be doped or undoped. The channel layer **106** may include a Group III-V compound, such as, but not limited to, GaAs, indium gallium arsenide (InGaAs), or GaN. In some embodiments, the channel layer **106** may include the same material as the

substrate **102**. Alternatively, in other embodiments, the channel layer **106** may include a material different than the material in the substrate **102**.

The barrier layer **108** has a band gap that is different from the band gap of the channel layer **106**. The barrier layer **108** may include one or more sub-layers and may be doped or undoped. The barrier layer **108** may include a Group III-V compound, such as, but not limited to, aluminum gallium arsenide (AlGaAs), indium phosphide (InP), or aluminum gallium nitride (AlGaN).

A heterojunction may be formed at the interface between the barrier layer **108** and the channel layer **106**, resulting in the formation of a two-dimensional electron gas (2DEG) (not shown) in the channel layer **106**. The 2DEG can provide the conductive pathway for the drifting of charges between the source electrode **120** and the drain electrode **118**. For example, the barrier layer **108** and the channel layer **106** may be substantially undoped, and the 2DEG may be induced by polarization mismatches between the barrier layer **108** and the channel layer **106**. In this example, the barrier layer **108** and the channel layer **106** may both include Group III-Nitride compounds, such as, but not limited to, GaN, AlN, and AlGaN. In another example, the barrier layer **108** may be more highly doped than the channel layer **106**, which forms the 2DEG.

The device may also include a buffer layer **104** between the channel layer **106** and the substrate **102**. In particular, the buffer layer **104** may be disposed on the substrate **102**. The buffer layer **104** may include group III or group V elements, such as, but are not limited to, AlN, GaN, or AlGaN. The buffer layer **104** may serve to accommodate the lattice mismatch between the channel layer **106** and the substrate **102**, and thereby improving the “on” state and “off” state electrical characteristics of the device while accommodating bowing, warping, breakage, or cracks of the substrate **102**. The buffer layer **104** may be a superlattice structure with multiple sub-layers, or alternatively, be structured as a single layer. The term “buffer layer” may alternatively be referred to as a “transitional layer”.

The device may further include an isolation layer **110** disposed on the buffer layer **104**. The isolation layer **110** may be laterally adjacent to channel layer **106**. The barrier layer **108** may be disposed on the isolation layer **110** such that the isolation layer **110** is positioned between the barrier layer **108** and the buffer layer **104**. In some embodiments, the isolation layer **110** may be a mesa-isolated layer or an implant-isolated layer. In other embodiments (not shown), the isolation layer **110** may be absent.

The source electrode **120** and the drain electrode **118** may contact the active region **150** in various configurations. For example, the gate **112** may be disposed on the barrier layer **108**. The source electrode **120** and the drain electrode **118** may contact or directly contact the channel layer **106** through the barrier layer **108** such that an electric current may flow between the source electrode **120** and the drain electrode **118** when the gate **112** is biased by a voltage. In other embodiments (not shown), the source electrode **120** and the drain electrode **118** may contact or directly contact the barrier layer **108** and form an electrical connection with the channel layer **106** through a diffusion-based thermal process.

An interlayer dielectric region **134** may be disposed over the gate **112**, the source electrode **120**, and the drain electrode **118**. The interlayer dielectric region **134** may include one or more dielectric layers. The dielectric layers in the interlayer dielectric region **134** may include silicon nitride, silicon dioxide, tetraethyl orthosilicate (TEOS), or a material

having a chemical composition of $\text{SiC}_x\text{O}_y\text{H}_z$, wherein x, y, and z are in stoichiometric ratio.

The polysilicon layer 122 may be disposed in the interlayer dielectric region 134 and may be positioned above the buffer layer 104. The polysilicon layer 122 may be preferably positioned laterally adjacent to the gate 112. The polysilicon layer 122 may be doped with dopants to have at least one doped region. For example, the polysilicon layer 122 may have a doped region with either a P-type conductivity or an N-type conductivity. Exemplary dopants for N-type conductivity may include, but are not limited to, arsenic, phosphorus, or antimony. Exemplary dopants for P-type conductivity may include, but are not limited to, boron, aluminum, or gallium. The polysilicon layer 122 may be disposed in the interlayer dielectric region 134 and the polysilicon layer 122 is also above the substrate 102. In embodiments where the isolation layer 110 is present, the polysilicon layer 122 may be positioned above or directly above the isolation layer 110 such that the polysilicon layer 122 is vertically aligned with the isolation layer 110. In embodiments where the isolation layer 110 is absent, the polysilicon layer 122 may be positioned above or directly above the active region 150. In particular, the polysilicon layer 122 may be positioned above or directly above the barrier layer 108.

A silicide layer 124 is disposed on the polysilicon layer 122. For example, the silicide layer 124 may be disposed on an upper surface of the polysilicon layer 122. The silicide layer 124 may include a metal silicide material. Examples of the metal silicide material for the silicide layer 124 may include, but are not limited to, nickel silicide, nickel-platinum silicide, cobalt silicide, titanium silicide. The silicide layer 124 may have a thickness in the range of about 5 nm to about 60 nm.

The polysilicon layer 122 may be positioned laterally adjacent to the gate 112 in various configurations. The polysilicon layer 122 may have a lower surface 123. The lower surface 123 of the polysilicon layer 122 may be substantially coplanar with, or at a lower elevation than, the upper surface 117 of the gate 112. For example, as shown in FIG. 1, the lower surface 123 of the polysilicon layer 122 may be substantially coplanar with the upper surface 117 of the gate 112. In examples shown in FIG. 2, FIG. 3, FIG. 4, the lower surface 123 of the polysilicon layer 122 may be at a lower elevation than the upper surface 117 of the gate 112. As shown in FIG. 2, the silicide layer 124 has an upper surface 125. The upper surface 125 of the silicide layer 124 may be substantially coplanar with, or at a lower elevation than, the lower surfaces 116 of the field plate extensions 114a, 114b. As shown in FIG. 3, the lower surface 123 of the polysilicon layer 122 may be substantially coplanar with, or at a lower elevation than, the lower surfaces 116 of the field plate extensions 114a, 114b. Spacers 126 may be disposed on sidewalls of the polysilicon layer 122. The spacers 126 may include a nitride material, such as silicon nitride.

In the example shown in FIG. 4, the spacers 126 may be absent from the sidewalls of the polysilicon layer 122. In particular, the silicide layer 124 may extend to lie over the sidewalls of the polysilicon layer 122. The upper surface 125 of the silicide layer 124 may be substantially coplanar with, or at a lower elevation than, the lower surfaces 116 of the field plate extensions 114a, 114b. Alternatively (not shown), the lower surface 123 of the polysilicon layer 122 may be substantially coplanar with, or at a lower elevation than, the lower surfaces 116 of the field plate extensions 114a, 114b.

Various interconnect features, such as interconnect vias 130a, 130b, 130c, 130d, 130e, and metal lines 132a, 132b,

132c, 132d, may be disposed in the interlayer dielectric region 134. As shown, interconnect vias 130a, 130b contact the silicide layer 124, interconnect via 130c contacts the drain electrode 118, interconnect via 130d contacts the gate 112, and interconnect via 130e contacts the source electrode 120. Each metal line 132a, 132b, 132c, 132d may be connected to at least one interconnect via 130a, 130b, 130c, 130d, 130e. The electrical connection between the drain electrode 118 and the polysilicon layer 122 may be provided by a connection through the silicide layer 124, interconnect vias 130b, 130c, and the metal line 132b.

Advantageously, by positioning the polysilicon layer 122 laterally adjacent to the gate 112, the polysilicon layer 122 can be positioned in the front end of line (FEOL) portion of an integrated circuit chip. By positioning the polysilicon layer 122 in the FEOL portion of the integrated circuit chip, the silicide layer 124 can be formed on the polysilicon layer 122. The presence of the silicide layer 124 may provide a lower contact resistance between the interconnect vias 130b, 130c and the polysilicon layer 122. Additionally, the presence of the silicide layer 124 may enable the polysilicon layer 122 to be configured as an electronic fuse. On the other hand, if the polysilicon layer 122 had been positioned in the back end of line (BEOL) portion of an integrated circuit chip (e.g., at a level above the metal lines 132a, 132b, 132c, 132d), it may not be possible to form a silicide layer on the polysilicon layer 122 without damaging other device components in the BEOL portion of the integrated circuit chip, since the formation of the silicide layer (i.e., a silicidation process) on the polysilicon layer 122 may include an annealing step.

In some embodiments, the polysilicon layer 122 may be configured as an electronic fuse. In embodiments where the polysilicon layer 122 is configured as an electronic fuse, the polysilicon layer 122 may be doped or undoped. The gate 112 may be configured to program the electronic fuse. For example, a pulse voltage may be applied to turn on the gate 112 and allow a current flow between the source electrode 120 and the drain electrode 118. The current flow from the drain electrode 118 may be sufficient to break the electronic fuse in the polysilicon layer 122. In other embodiments, the polysilicon layer 122 may be configured as a resistor. In embodiments where the polysilicon layer 122 is configured as a resistor, the polysilicon layer 122 may be doped.

The semiconductor devices described herein may include a HEMT device. Examples of a HEMT may include, but are not limited to, an aluminum gallium nitride (AlGaIn)/gallium nitride (GaN) HEMT, a pseudomorphic HEMT (PHEMT) (e.g., a gallium arsenide (GaAs) PHEMT), a metamorphic HEMT (mHEMT) (e.g., a GaAs mHEMT), a metal insulator semiconductor high electron mobility transistor (MISHEMT). The polysilicon layer 122 can be integrated into a HEMT device by positioning the polysilicon layer 122 above the barrier layer 108, the channel layer 106, the buffer layer 104, and the substrate 102.

Referring to FIG. 5, in which like reference numerals refer to like features in FIG. 1 through FIG. 4, another example of a semiconductor device is shown. In the example shown in FIG. 5, the polysilicon layer 122 may include a first doped region 122a and a second doped region 122b. In embodiments where the isolation layer 110 is present, both the first doped region 122a and the second doped region 122b may be positioned above or directly above the isolation layer 110 such that the doped regions 122a, 122b are vertically aligned with the isolation layer 110.

The first doped region 122a has a first conductivity type while the second doped region 122b has a second conduc-

tivity type. The first conductivity type is different from the second conductivity type. In particular, the first conductivity type may be of an opposite conductivity to the second conductivity type. For example, the first doped region **122a** may be of a P-type conductivity while the second doped region **122b** may be of an N-type conductivity type. Alternatively, the first doped region **122a** may be of an N-type conductivity while the second doped region **122b** may be of a P-type conductivity type. In some embodiments, the first doped region **122a** may abut or directly abut the second doped region **122b** to provide a diode in the polysilicon layer **122**.

The polysilicon layer **122** may have an upper surface that is at least partially covered by a silicide layer. For example, as shown in FIG. 5, a first silicide layer **124a** may be disposed on or directly on the first doped region **122a** of the polysilicon layer **122**, and a second silicide layer **124b** may be disposed on or directly on the second doped region **122b** of the polysilicon layer **122**. A nitride block **127** may be disposed on the polysilicon layer **122** to isolate the first silicide layer **124a** from the second silicide layer **124b**, thereby providing a partially silicided polysilicon layer **122**. The first doped region **122a** may be electrically connected to the drain electrode **118**. For example, the electrical connection may be provided by a connection between the first silicide layer **124a**, the interconnect vias **130b**, **130c**, and the metal line **132b**.

Referring to FIG. 6, in which like reference numerals refer to like features in FIG. 1 through FIG. 5, another example of a semiconductor device is shown. The polysilicon layer **122** may include heavily doped regions and lightly doped regions. In the example shown in FIG. 6, the polysilicon layer **122** may further include a first heavily doped region **136**, a second heavily doped region **138**, a third heavily doped region **140**, a first lightly doped region **137**, and a second lightly doped region **139**. The heavily doped regions **136**, **138**, **140** have higher dopant concentrations than the lightly doped regions **137**, **139**. Each lightly doped region **137**, **139** may be positioned laterally between two respective heavily doped regions **136**, **138**, **140**. The lightly doped regions **137**, **139** may be intrinsically doped. As used herein, the term “intrinsically doped” may refer to a region that is substantially free of extrinsic doping atoms but may still contain charge carriers provided by other doped portions of the structure.

The second heavily doped region **138** may be positioned laterally between the first heavily doped region **136** and the third heavily doped region **140**. The second heavily doped region **138** may include sub-regions **138a**, **138b**. Sub-region **138a** may be laterally adjacent to sub-region **138b**. The second heavily doped region **138** may include a diode. For example, the sub-region **138a** may be doped with a conductivity type that is opposite of the sub-region **138b**. The sub-region **138a** may abut or directly abut the sub-region **138b** to provide the diode.

The heavily doped region **136** may be of a first conductivity type. The sub-region **138a** of the heavily doped region **138** may be of a second conductivity type while the sub-region **138b** of the heavily doped region **138** may be of the first conductivity type. The heavily doped region **140** may be of the second conductivity type. The first conductivity type is different from the second conductivity type. In particular, the first conductivity type may be of an opposite conductivity to the second conductivity type.

A silicide layer may be disposed on or directly on each of the heavily doped regions **136**, **138**, **140**. For example, a first silicide layer **124a** may be disposed on the heavily doped

region **136**, a second silicide layer **124b** may be disposed on the heavily doped region **138**, and a third silicide layer **124c** may be disposed on the heavily doped region **140**. Nitride blocks **127** may be disposed on the lightly doped regions **137**, **139**. The silicide layers **124a**, **124b**, **124c** may be isolated from each other by the nitride blocks **127**. The heavily doped region **140** may be electrically connected to the drain electrode **118**, for example, by interconnect vias **130c**, **130d**, the metal line **132c**, and the first silicide layer **124a**. In an embodiment, the heavily doped region **138** may be electrically connected to a heat sink.

Although not shown, it should be understood that the scope of the present disclosure also contemplates modifications of the structures shown in FIG. 5 and FIG. 6 where the isolation layer **110** is absent. Additionally, the present disclosure also contemplates modifications of the structures shown in FIG. 1 through FIG. 6 where the polysilicon layer **122** and the respective doped regions in the polysilicon layer **122** are electrically connected to the source electrode **120**, or the gate **112**.

FIG. 7, FIG. 8, FIG. 9A, FIG. 9B, FIG. 10 through FIG. 16 show example structures associated with steps that may be used to create an example of the semiconductor device shown in FIG. 2.

As used herein, “deposition techniques” refer to the process of applying a material over another material (or the substrate). Exemplary techniques for deposition include, but are not limited to, spin-on coating, sputtering, chemical vapor deposition (CVD), physical vapor deposition (PVD), molecular beam deposition (MBD), pulsed laser deposition (PLD), liquid source misted chemical deposition (LSMCD), atomic layer deposition (ALD).

Additionally, “patterning techniques” include deposition of material or photoresist, patterning, exposure, development, etching, cleaning, and/or removal of the material or photoresist as required in forming a described pattern, structure, or opening. Examples of techniques for patterning include, but are not limited to, wet etch lithographic processes, dry etch lithographic processes, or direct patterning processes. Such techniques may use mask sets and mask layers.

Referring to FIG. 7, a device structure for use in fabricating the semiconductor devices of the present disclosure is shown. A buffer layer **104** may be formed on a substrate **102**. The substrate **102** may be provided as a semiconductor wafer. An active region **150** may be formed above a substrate **102**. For example, a channel layer **106** may be formed on the buffer layer **104**, and a barrier layer **108** may be formed on the channel layer **106**. A first dielectric layer **134a** may be formed on the barrier layer **108** using the deposition techniques described herein. In some embodiments, an isolation layer **110** may be formed between the barrier layer **108** and the buffer layer **104**. The isolation layer **110** may be implant-isolated or mesa-isolated. In other embodiments, the forming of the isolation layer **110** may be skipped (i.e., the isolation layer **110** is absent), and the channel layer **106** is between the barrier layer **108** and the buffer layer **104**.

Referring to FIG. 8, a polysilicon material **141** may be formed on the first dielectric layer **134a** using the deposition techniques described herein. The deposited polysilicon material **141** may be undoped. Alternatively, a doped polysilicon material **141** may be deposited on the first dielectric layer **134a**. The doped polysilicon material **141** may be doped with either a P-type conductivity or an N-type conductivity.

Referring to FIG. 9A, the polysilicon material **141** may be patterned to form a polysilicon layer **122** using the pattern-

ing techniques described herein. In embodiments where the polysilicon material **141** was undoped, a doping step may be performed to dope the polysilicon layer **122** to have a doped region. The doping of the polysilicon layer **122** may be performed with the use of a mask (not shown). Spacers **126** may be formed on sidewalls of the polysilicon layer **122**. For example, a nitride material may be deposited and subsequently anisotropically etched to form the spacers **126**. Alternatively, to form the embodiment shown in FIG. 4, the formation of the spacers **126** on the sidewalls of the polysilicon layer **122** may be omitted, thereby leaving the sidewalls of the polysilicon layer **122** exposed.

Referring to FIG. 9B, a silicide layer **124** may be formed on the polysilicon layer **122** using a silicidation process. The spacers **126** may serve as a mask to block surfaces of the polysilicon layer where silicidation is not desired. For example, the sidewalls of the polysilicon layer **122** may be covered by the spacers **126** while an upper surface of the polysilicon layer **122** may be exposed. A metallic material (e.g., nickel, nickel-platinum alloy, titanium, or cobalt, etc.) may be formed on the upper surface of the polysilicon layer **122** using a sputtering process. An annealing process may be performed to form the silicide layer **124** on the upper surface of the polysilicon layer **122** from a reaction between the metallic material and the polysilicon. The annealing process may be conducted with a temperature range of about 250° C. to about 750° C. Alternatively, to form the embodiment shown in FIG. 4, the metallic material may be sputtered on the exposed sidewalls and the upper surface of the polysilicon layer **122**. The annealing process may be subsequently performed to form the silicide layer **124**. In particular, in the absence of spacers **126** blocking the sidewalls of the polysilicon layer **122**, the silicide layer **124** may be formed on the upper surface and the sidewalls of the polysilicon layer **122**.

Referring to FIG. 10, a second dielectric layer **134b** may be formed on the first dielectric layer **134a** using the deposition techniques described herein. For example, the second dielectric layer **134b** may be deposited to cover the silicide layer **124** and the polysilicon layer **122**. A gate opening **142** may be formed in the dielectric layers **134a**, **134b** by an etching process. In particular, the gate opening **142** may expose the barrier layer **108**.

Referring to FIG. 11, a gate **112** may be formed in the gate opening **142**. The gate **112** may be formed above the active region **150**. In an embodiment, the gate **112** may be formed on the barrier layer **108**. The gate **112** may be formed by depositing a metallic material layer (not shown) in the gate opening **142** using the deposition techniques described herein. The deposited layer may be in contact with the barrier layer **108** and extend to lie over the second dielectric layer **134b**. The deposited layer may be subsequently patterned using the patterning techniques described herein, with the use of a mask, to form the gate **112**. The gate **112** may also have field plate extensions **114a**, **114b**. The field plate extensions **114a**, **114b** may be integrally formed with the gate **112** after the patterning of the metallic material layer. The field plate extensions **114a**, **114b** may be formed on the second dielectric layer **134b**.

Referring to FIG. 12, a third dielectric layer **134c** may be formed on the second dielectric layer **134b** using the deposition techniques described herein. Source **144a** and drain **144b** openings may be formed in the dielectric layers **134a**, **134b**, **134c** by an etching process. A mask may be used in the etching process to define the openings **144a**, **144b** in the dielectric layers **134a**, **134b**, **134c**. The etching process may also etch through the barrier layer **108** to expose the channel layer **106**.

Referring to FIG. 13, an electrode layer **152** may be formed on the third dielectric layer **134c** and in the openings **144a**, **144b**, using the deposition techniques described herein. Preferably, the electrode layer **152** may be formed using a conformal deposition process, such as an ALD process or a highly conformal CVD process. As used herein, the term “conformal” refers to a deposition technique in which the deposited material conforms to the contours of the surface or structure being deposited on. The conformal deposition process may provide the electrode layer **152** with a uniform thickness. As shown in FIG. 13, the deposited electrode layer **152** may conform to sidewalls of the openings **144a**, **144b** and contact the channel layer **106**.

Referring to FIG. 14, the electrode layer **152** may be patterned using the patterning techniques described herein to form a source electrode **120** and a drain electrode **118**. For example, a mask element **146** may be deposited in each of the openings **144a**, **144b**. The mask element **146** may fill the openings **144a**, **144b** and cover portions of the electrode layer **152**. Portions of the electrode layer **152** not covered by the mask elements **146** may be removed using the patterning techniques described herein (e.g., an etching process). In some examples, the etching of the electrode layer **152** may leave behind electrode spacers **128** on the third dielectric layer **134c**, in which the gate **112** may be laterally between the electrode spacers **128**. In other examples, the etching of the electrode layer **152** does not leave behind the electrode spacers **128**.

Referring to FIG. 15, the mask element **146** may be removed (e.g., using an ashing step) after the formation of the source electrode **120** and the drain electrode **118**. The removal of the mask element **146** may expose the source electrode **120** and the drain electrode **118**. A fourth dielectric layer **134d** may be formed on the third dielectric layer **134c**, the source electrode **120**, and the drain electrode **118**.

Referring to FIG. 16, contact openings **148** may be formed in the dielectric layers **134a**, **134b**, **134c**, **134d**. For example, the dielectric layers **134a**, **134b**, **134c**, **134d** may be patterned by performing an etching process with the use of a mask. The contact openings **148** may expose portions of the silicide layer **124**, the source electrode **120**, the drain electrode **118**, and the gate **112**.

The structure shown in FIG. 16 may be subjected to further processing. For example, interconnect vias may be formed in the contact openings **148** (e.g., using a damascene process, or deposition of metals). A chemical mechanical planarization (CMP) step may be performed thereafter to planarize the upper surfaces of the interconnect vias. Metal lines may be formed on the planarized interconnect vias. The dielectric layers **134a**, **134b**, **134c**, **134d** may provide an interlayer dielectric region, in which the interconnect vias and the metal lines may be disposed in.

FIG. 17A and FIG. 17B show structures at stages subsequent to FIG. 8 for forming a semiconductor device having a polysilicon layer including a first doped region and a second doped region. Referring to FIG. 17A (FIG. 17A continues from the structure shown in FIG. 8), the polysilicon material **141** may be patterned to form a polysilicon layer **122** using the patterning techniques described herein. In embodiments where the polysilicon material **141** was undoped, a doping step may be performed to dope the polysilicon layer **122** to have a first doped region **122a** and a second doped region **122b**. A mask may be used in the doping step. The first doped region **122a** may be doped with a first conductivity type while the second doped region **122b** may be doped with a second conductivity type. In embodiments where the polysilicon material **141** was doped, an

additional doped polysilicon material may be deposited adjacent to the patterned polysilicon material **141** to form the first doped region **122a** of the polysilicon layer **122** and the second doped region **122b** of the polysilicon layer **122**.

Spacers **126** may be formed on sidewalls of the polysilicon layer **122**. A nitride block **127** may be formed on an upper surface of the polysilicon layer **122**. The spacers **126** and the nitride block **127** may be formed by depositing and subsequently etching a nitride material. The upper surface of the polysilicon layer **122** may be partially covered by the nitride block **127** such that an upper surface of the first doped region **122a** and an upper surface of the second doped region **122b** are exposed.

Referring to FIG. **17B**, a first silicide layer **124a** may be formed on the first doped region **122a** of the polysilicon layer **122**, and a second silicide layer **124b** may be formed on the second doped region **122b** of the polysilicon layer **122**. The first silicide layer **124a** may be isolated from the second silicide layer **124b** by the nitride block **127**.

Formation of the first silicide layer **124a** and the second silicide layer **124b** may be performed using a silicidation process. The spacers **126** and the nitride block **127** may serve as masks to block surfaces of the polysilicon layer **122** where silicidation is not desired. A metallic material (e.g., nickel, nickel-platinum alloy, titanium, or cobalt, etc.) may be formed (e.g., using a sputtering process) on the exposed upper surface of the first doped region **122a** and the exposed upper surface of the second doped region **122b**. An annealing process may be performed to form the first silicide layer **124a** and the second silicide layer **124b** from a reaction between the metallic material and the polysilicon. The annealing process may be conducted with a temperature range of about 350° C. to about 750° C. Subsequently, the structure shown in FIG. **17B** may be subjected to similar processes described in FIG. **10** through FIG. **16** to form the exemplary semiconductor device shown in FIG. **5**.

FIG. **18A** and FIG. **18B** show structures at stages subsequent to FIG. **8** for forming a semiconductor device having a polysilicon layer including heavily and lightly doped regions. Referring to FIG. **18A** (FIG. **18A** continues from the structure shown in FIG. **8**), the polysilicon material **141** may be patterned, using the patterning techniques described herein, to form a polysilicon layer **122**. The polysilicon layer **122** may be doped to have heavily doped regions **136**, **138**, **140** and lightly doped regions **137**, **139**.

In an example, the polysilicon layer **122** may be doped to have a first heavily doped region **136** of a first conductivity type, a second heavily doped region **138**, a third heavily doped region **140** of a second conductivity type, a first lightly doped region **137**, and a second lightly doped region **139**. The first lightly doped region **137** may be formed laterally between the first heavily doped region **136** and the second heavily doped region **138**. The second lightly doped region **139** may be formed laterally between the second heavily doped region **138** and the third heavily doped region **140**. The second heavily doped region **138** may additionally include a first sub-region **138a** and a second sub-region **138b**. The first sub-region **138a** may be of the first conductivity type while the second sub-region **138b** may be of the second conductivity type.

The forming of the doped regions in the polysilicon layer **122** may involve one or more doping steps (e.g., ion implantation) with the use of one or more masks, and a heating step (e.g., annealing). As an example, the heavily doped regions **136**, **138**, **140** may be formed by ion implantation. An anneal process may be performed after the ion implantation. The lightly doped regions **137**, **139** may be

formed by a thermal diffusion of the dopants from the respective adjacent heavily doped regions **136**, **138**, **140**.

Spacers **126** may be formed on sidewalls of the polysilicon layer **122**. Nitride blocks **127** may be formed on the lightly doped regions **137**, **139**. The nitride blocks **127** may also extend to partially cover the upper surfaces of the respective heavily doped regions **136**, **138**, **140**. The spacers **126** and the nitride blocks **127** may be formed by depositing and subsequently etching a nitride material.

Referring to FIG. **18B**, a first silicide layer **124a** may be formed on the first heavily doped region **136**, a second silicide layer **124b** may be formed on the second heavily doped region **138**, and a third silicide layer **124c** may be formed on the third heavily doped region **140**. The nitride blocks **127** may serve to isolate the first silicide layer **124a** from the third silicide layer **124c**, and the second silicide layer **124b** from the third silicide layer **124c**.

Formation of the first silicide layer **124a**, the second silicide layer **124b**, and the third silicide layer **124c** may be performed using a silicidation process. The spacers **126** and the nitride block **127** may serve as masks to block surfaces of the doped regions **136**, **137**, **138**, **139**, **140** in the polysilicon layer **122** where silicidation is not desired. A metallic material (e.g., nickel, nickel-platinum alloy, titanium, or cobalt, etc.) may be formed (e.g., using a sputtering process) on the exposed upper surface of the first heavily doped region **136**, the exposed upper surface of the second heavily doped region **138**, and the exposed upper surface of the third heavily doped region **140**. An annealing process may be performed to form the silicide layers **124a**, **124b**, **124c** from a reaction between the metallic material and the polysilicon. The annealing process may be conducted with a temperature range of about 350° C. to about 750° C. Subsequently, the structure shown in FIG. **18B** may be subjected to similar processes described in FIG. **10** through FIG. **16** to form the exemplary semiconductor device shown in FIG. **6**.

Throughout this disclosure, it is to be understood that if a method is described herein as involving a series of steps, the order of such steps as presented herein is not necessarily the only order in which such steps may be performed, and certain of the stated steps may possibly be omitted and/or certain other steps not described herein may possibly be added to the method. Furthermore, the terms “comprise”, “include”, “have”, and any variations thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or device that comprises a list of elements is not necessarily limited to those elements, but may include other elements not expressly listed or inherent to such process, method, article, or device. Occurrences of the phrase “in an embodiment” herein do not necessarily all refer to the same embodiment.

The descriptions of the various embodiments of the present disclosure have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein. Furthermore, there is no intention to be bound by any theory presented in the preceding background or the following detailed description. Additionally, the various tasks and processes described herein may be incorpo-

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rated into a more comprehensive procedure or process having additional functionality not described in detail herein.

References herein to terms modified by language of approximation, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. The language of approximation may correspond to the precision of an instrument used to measure the value and, unless otherwise dependent on the precision of the instrument, may indicate $\pm 10\%$ of the stated value(s).

As will be readily apparent to those skilled in the art upon a complete reading of the present application, the disclosed semiconductor devices and methods of forming the same may be employed in manufacturing a variety of different integrated circuit products, including, but not limited to, high power electronic devices, communication devices, high electron mobility transistors, etc.

What is claimed is:

1. A semiconductor device comprising:
 - an active region above a substrate, the active region comprising at least two material layers with different band gaps, wherein the active region comprises a channel layer and a barrier layer disposed on the channel layer;
 - source and drain electrodes in contact with the active region;
 - a gate above the active region, the gate is laterally between the source and drain electrodes, the gate has an upper surface;
 - a polysilicon layer above the substrate, the polysilicon layer has a lower surface, wherein the polysilicon layer is positioned laterally adjacent to the gate and the lower surface of the polysilicon layer is substantially coplanar with the upper surface of the gate; and
 - a silicide layer on the polysilicon layer.
2. The device of claim 1, further comprising an isolation layer above the substrate, the isolation layer being adjacent to the channel layer.
3. The device of claim 2, wherein the polysilicon layer is positioned above the isolation layer.
4. The device of claim 3, wherein the polysilicon layer is vertically aligned with the isolation layer.
5. The device of claim 1, wherein the gate comprises field plate extensions that extend laterally from the gate towards the source and drain electrodes.
6. The device of claim 1, further comprising:
 - a buffer layer disposed on the substrate, wherein the channel layer is disposed on the buffer layer, and the isolation layer disposed on the buffer layer.
7. The device of claim 1, wherein the polysilicon layer is electrically connected to either the drain electrode or the source electrode.
8. The device of claim 1, wherein the polysilicon layer includes at least one doped region, the doped region having a conductivity type.
9. The device of claim 8, wherein the polysilicon layer includes a first doped region with a first conductivity type and a second doped region with a second conductivity type, the first conductivity type being different from the second conductivity type.
10. The device of claim 9, wherein first doped region abuts the second doped region to provide a diode.
11. The device of claim 7, wherein the polysilicon layer is configured as an electronic fuse.

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12. The device of claim 11, wherein the gate is configured to program the electronic fuse.

13. The device of claim 1, wherein the polysilicon layer is configured as a resistor.

14. A semiconductor device comprising:

- a buffer layer on a substrate;
- an active region above the substrate, the active region comprising a channel layer disposed on the buffer layer, and a barrier layer disposed on the channel layer;
- source and drain electrodes in contact with the channel layer;
- a gate on the barrier layer, the gate is laterally between the source and drain electrodes, the gate has an upper surface;
- a polysilicon layer above the barrier layer, the polysilicon layer has a lower surface, the polysilicon layer is positioned laterally adjacent to the gate, and the lower surface of the polysilicon layer is substantially coplanar with the upper surface of the gate, wherein the polysilicon layer comprises heavily doped regions and lightly doped regions; and
- a silicide layer on each of the heavily doped regions.

15. The device of claim 14, wherein each of the lightly doped regions is positioned laterally between two heavily doped regions.

16. The device of claim 15, wherein the polysilicon layer comprises:

- a first heavily doped region of a first conductivity type;
- a second heavily doped region having a first sub-region and a second sub-region, wherein the first sub-region is of a second conductivity type and the second sub-region is of the first conductivity type, the second conductivity type being different from the first conductivity type;
- a third heavily doped region of the second conductivity type;
- a first lightly doped region positioned laterally between the first heavily doped region and the second heavily doped region; and
- a second lightly doped region positioned laterally between the second heavily doped region and the third heavily doped region.

17. The device of claim 16, wherein the first sub-region of the second heavily doped region abuts the second sub-region of the second heavily doped region.

18. The device of claim 17, further comprising:

- a first silicide layer on the first heavily doped region;
- a second silicide layer on the second heavily doped region; and
- a third silicide layer on the third heavily doped region.

19. A method of forming a semiconductor device comprising:

- forming an active region above a substrate, the active region comprising at least two material layers with different band gaps, wherein the active region comprises a channel layer and a barrier layer disposed on the channel layer;
- forming source and drain electrodes to contact the active region;
- forming a gate above the active region, the gate is laterally between the source and drain electrodes, the gate has an upper surface;

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forming a polysilicon layer above the substrate, the polysilicon layer has a lower surface, wherein the polysilicon layer is positioned laterally adjacent to the gate and the lower surface of the polysilicon layer is substantially coplanar with the upper surface of the gate; and 5
forming a silicide layer on the polysilicon layer.

* * * * *

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